

Olympiad Test Paper — Mathematics (Class 7)

Chapter 1: Large Numbers Around Us

Subject	Mathematics (NCERT Class 7)
Chapter	1 — Large Numbers Around Us
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Work in the Indian system (3-2-2-2 commas: lakh, crore, arab) unless a question names the International system (3-3-3: thousand, million, billion). Watch the classic traps — confusing one lakh with one million, miscounting zeros, putting commas in the wrong groups, and rounding the wrong way. Do not write on this paper — mark answers on your answer sheet.

1. How many zeroes follow the leading 1 in one crore?

- (A) 5 (B) 6
(C) 7 (D) 8

2. Written with Indian commas, the number 50000000 is:

- (A) 5,00,00,000 (B) 50,00,000
(C) 5,0000,000 (D) 50,000,000

3. One million equals how many lakhs?

- (A) 1 (B) 1000
(C) 100 (D) 10

4. The largest 5-digit number is 99,999. The number that comes just after it is:

- (A) 10,000 (B) 1,00,000
(C) 9,99,999 (D) 1,00,001

5. In the number 7,84,36,512 the digit 4 stands for:

- (A) 4 lakh (B) 40 lakh
(C) 4 thousand (D) 40 thousand

6. The number name "three crore five lakh seven thousand two hundred" in figures is:

- (A) 30,57,200
(C) 3,05,70,200

- (B) 3,50,07,200
(D) 3,05,07,200

7. The same number is written 9,07,06,005 in the Indian system. In the International system it is:

- (A) 9,070,6005
(C) 907,06,005

- (B) 90,706,005
(D) 90,76,005

8. What is the smallest 8-digit number?

- (A) 10,00,000
(C) 1,00,00,000

- (B) 10,00,00,000
(D) 99,99,999

9. The place value of 6 minus the place value of 3 in the number 6,53,000 is:

- (A) 5,97,000
(C) 5,70,000

- (B) 3,00,000
(D) 5,99,700

10. Reading right to left, the Indian comma convention groups the digits as:

- (A) 3, then 3, then 3, ...
(C) 2, then 3, then 3, ...

- (B) 2, then 2, then 2, ...
(D) 3, then 2, then 2, then 2, ...

11. How many lakhs make one arab?

- (A) 100
(C) 10,000

- (B) 1,000
(D) 1,00,000

12. Estu writes "one billion = one hundred lakh." His teacher marks it wrong. The correct value of one billion in lakhs is:

- (A) 100 lakh
(C) 1,000 lakh

- (B) 10,000 lakh
(D) 1,00,000 lakh

13. Which of these is the smallest number?

- (A) 9 lakh
(C) 90 lakh

- (B) 1 million
(D) 9,00,001

14. The expanded form $4 \times 10,00,000 + 7 \times 10,000 + 3 \times 100 + 5$ represents:

- (A) 4,07,305
(C) 40,07,305

- (B) 40,70,305
(D) 47,00,305

15. To display 53,000 using only a +1000 button (starting from 0), the number of presses needed is:

- (A) 5.3
(C) 5,300

- (B) 53
(D) 530

16. Handy Hundreds has only a +100 button. Starting from 0, how many presses reach one lakh?

- (A) 1,000 (B) 100
(C) 10,000 (D) 100,00

17. Systematic Sippy makes 40,629 with the fewest presses of +1, +10, +100, +1000, +10000. The total number of presses is:

- (A) 19 (B) 20
(C) 21 (D) 22

18. Creative Chitti reaches 8,300 using +1000 seven times and the +100 button some number of times. How many times is +100 pressed?

- (A) 3 (B) 8
(C) 83 (D) 13

19. If Systematic Sippy uses the fewest presses, the press-counts in each slot for the number 3,07,250 (using +1 ... +1,00,000) are:

- (A) 3, 0, 7, 2, 5, 0 (B) 3, 7, 2, 5, 0
(C) 3, 0, 7, 2, 0, 5, 0 (D) 3, 0, 0, 7, 2, 5, 0

20. Which press-recipe does NOT make 5,072?

- (A) $5 \times 1000 + 7 \times 10 + 2 \times 1$ (B) $50 \times 100 + 72 \times 1$
(C) $4 \times 1000 + 10 \times 100 + 70 \times 1$ (D) $5 \times 1000 + 6 \times 10 + 12 \times 1$

21. The number 12,78,830 read in words is:

- (A) one crore twenty-seven lakh eighty-eight thousand thirty
(B) one lakh twenty-seven thousand eight hundred thirty
(C) twelve lakh seventy-eight thousand eight hundred three
(D) twelve lakh seventy-eight thousand eight hundred thirty

22. In 6,72,85,183, the digit in the ten-lakh place is:

- (A) 7 (B) 6
(C) 2 (D) 8

23. The difference between the place values of the two 5s in 5,35,000 is:

- (A) 4,65,000 (B) 5,00,000
(C) 4,95,000 (D) 4,99,500

24. "Ten crore twenty lakh thirty thousand forty," re-commaded in the International system, is:

- (A) 10,20,30,040 (B) 102,030,040
(C) 102,30,040 (D) 10,203,0040

25. Which statement about 10,00,000 (Indian) and 1,000,000 (International) is true?

- (A) The Indian one is ten times larger (B) They are the same number, ten lakh
(C) The International one is ten times larger (D) They are the same number, one lakh

- 26.** According to the 2011 Census, Chintamani's population was about 75,000. How much less than one lakh is that?
- (A) 25,000 (B) 15,000
(C) 35,000 (D) 1,25,000
- 27.** A factory made 4,87,650 toys. Rounded to the nearest ten-thousand, the report prints:
- (A) 4,80,000 (B) 4,88,000
(C) 5,00,000 (D) 4,90,000
- 28.** The nearest thousand for 3,87,69,957 is:
- (A) 3,87,69,000 (B) 3,87,69,900
(C) 3,87,70,000 (D) 3,88,00,000
- 29.** A principal must give one sweet to each of 732 students and 23 staff, and never run short. The most sensible order is:
- (A) round up to 800 (B) exactly 755
(C) round down to 700 (D) round down to 750
- 30.** A shopkeeper quotes a ₹470 item as "about ₹450." This is rounding:
- (A) up, to seem generous (B) down, to seem generous
(C) up, to the nearest ten (D) down, to the nearest hundred
- 31.** The nearest lakh for the number 6,72,85,183 is:
- (A) 6,72,00,000 (B) 6,72,85,000
(C) 6,73,00,000 (D) 6,72,90,000
- 32.** Roxie estimates $4,63,128 + 4,19,682$ as "near 8,00,000 and a bit more." The exact sum is:
- (A) 8,72,810 (B) 9,82,810
(C) 8,82,710 (D) 8,82,810
- 33.** Which of these is the best reason to report a fair's attendance as "about one lakh" rather than 1,02,455?
- (A) the round figure conveys the size with less effort and no real loss
(B) one lakh is larger and sounds better (C) the exact figure is wrong
(D) one lakh is the only number allowed for crowds
- 34.** Rounded to the nearest crore, the number 8,49,21,000 becomes:
- (A) 8,50,00,000 (B) 8,00,00,000
(C) 9,00,00,000 (D) 8,49,00,000
- 35.** A number rounds to 7,00,000 at the nearest lakh. Which of these could the number be?
- (A) 7,51,000 (B) 6,40,000
(C) 7,48,200 (D) 7,62,000

36. Using the " $\times 5$ means $\times 10$ then $\div 2$ " shortcut, 116×5 equals:

- (A) 480
- (B) 580
- (C) 1,160
- (D) 5,800

37. Compute 824×25 using " $\times 25$ means $\times 100$ then $\div 4$ ":

- (A) 2,060
- (B) 82,400
- (C) 8,240
- (D) 20,600

38. Compute 72×125 mentally ($125 = 1000 \div 8$):

- (A) 9,000
- (B) 900
- (C) 90,000
- (D) 576

39. Without multiplying it out, how many digits does the product 234×567 have?

- (A) 4
- (B) 5 only
- (C) 6 only
- (D) 5 or 6

40. The product of a 2-digit number and a 2-digit number can have:

- (A) exactly 4 digits
- (B) 3 or 4 digits
- (C) 2 or 3 digits
- (D) 4 or 5 digits

41. Multiplying a 3-digit number by a 3-digit number, the smallest possible product is:

- (A) 1,000
- (B) 9,801
- (C) 10,000
- (D) 99,800

42. How many digits does the product of a 12-digit number and a 13-digit number have?

- (A) 25 only
- (B) 156
- (C) 24 only
- (D) 24 or 25

43. Without multiplying exactly, the number of digits in $8,000 \times 7,000$ is:

- (A) 7
- (B) 8
- (C) 6
- (D) 9

44. Roxie claims "a 3-digit number times a 3-digit number can give a 4-digit answer." She is:

- (A) right; e.g. 100×10
- (B) right; e.g. 999×1
- (C) wrong; the smallest such product is 1,000
- (D) wrong; the smallest such product is 10,000

45. The shortcut $116 \times 5 = (116 \times 10) \div 2$ works because:

- (A) 5 is half of 10
- (B) 5 is double of 10
- (C) $\times 10$ always makes a number 5 times bigger
- (D) dividing by 2 is the same as multiplying by 2

46. Each bus holds about 50 people. About how many people fit in 1 lakh buses?

- (A) 5 lakh (B) 5 crore
(C) 50 lakh (D) 50,000

47. At one number per second, about how far could you count in a single day?

- (A) about one lakh (B) about 86,400
(C) about ten lakh (D) about 36,000

48. Roxie walks 100 km a day to reach the Moon (3,84,400 km away). Walking every day for 10 years (365 days each), she covers:

- (A) 36,500 km — far short (B) 4,00,000 km — overshoots
(C) 3,84,400 km — exactly arrives (D) 3,65,000 km — just short

49. 5,000 ships each carry about 2,500 passengers. Together they hold about:

- (A) 1,25,000 (B) 12,50,000
(C) 1,25,00,000 (D) 1,25,00,00,000

50. Ahaan's heart beats about 75 times a minute. Roughly how many beats in a full day?

- (A) about 10,000 (B) about 4,500
(C) about 10,00,000 (D) about 1,00,000

51. Which assumption is the most reasonable for a Fermi estimate?

- (A) one barbershop serves about 5,000 people in a locality
(B) one bus holds 5,000 people (C) one barbershop does 1,000 haircuts a day
(D) a city has exactly one barbershop

52. Compare 500 lakhs with 5 million. Which is true?

- (A) they are equal (B) 500 lakhs is bigger (it is 5 crore)
(C) 5 million is bigger
(D) 500 lakhs equals 50 million only after rounding

53. Compare 640 crore and 60 billion. The bigger one is:

- (A) 640 crore (it is 64 billion) (B) they are equal
(C) 60 billion (640 crore is only 6.4 billion) (D) 640 crore equals 640 billion

54. Arrange in increasing order: 9 lakh, 1 million, 90,000, 1 crore.

- (A) $90,000 < 9 \text{ lakh} < 1 \text{ million} < 1 \text{ crore}$ (B) $9 \text{ lakh} < 90,000 < 1 \text{ million} < 1 \text{ crore}$
(C) $90,000 < 1 \text{ million} < 9 \text{ lakh} < 1 \text{ crore}$ (D) $1 \text{ million} < 9 \text{ lakh} < 90,000 < 1 \text{ crore}$

55. A water tank's reading rose from 5,00,000 litres to one crore litres. The increase is:

- (A) 5,00,000 litres (B) 50,00,000 litres
(C) 95,00,000 litres (D) 9,50,00,000 litres

56. The smallest 9-digit number, and the number just before it, are:

- (A) 1,00,00,000 and 99,99,999 (B) 10,00,00,000 and 9,99,99,999
(C) 10,00,00,000 and 99,99,999 (D) 1,00,00,00,000 and 9,99,99,999

57. A district has 12 lakh people; the state it sits in has 8 crore. The district's people are what fraction of the state's?

(A) $1/8$

(B) $3/20$

(C) $1/80$

(D) $3/200$

58. Mumbai's population is over 1 crore 24 lakh. If 1 lakh buses each hold 50 people, those buses hold what fraction of Mumbai's people (roughly)?

(A) about 40%

(B) about all of them

(C) about 4%

(D) about 400%

59. Which comparison is correct?

(A) $7,05,000 > 70,50,000$

(B) $9,99,999 > 10,00,000$

(C) $1,00,01,000 > 99,99,000$

(D) 8 lakh > 8 million

60. A report rounds 49,873 visitors to the nearest ten-thousand and also reports them as "about a lakh." Which is true?

(A) nearest ten-thousand is 40,000; "about a lakh" is dishonest

(B) nearest ten-thousand is 60,000; "about a lakh" understates

(C) nearest ten-thousand is 50,000; "about a lakh" is exactly right

(D) nearest ten-thousand is 50,000; "about a lakh" overstates by roughly double

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 1: Large Numbers Around Us

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	D	31	C	41	C	51	A
2	A	12	B	22	A	32	D	42	D	52	B
3	D	13	A	23	C	33	A	43	B	53	C
4	B	14	B	24	B	34	B	44	D	54	A
5	A	15	B	25	B	35	C	45	A	55	C
6	D	16	A	26	A	36	B	46	C	56	B
7	B	17	C	27	D	37	D	47	B	57	D
8	C	18	D	28	C	38	A	48	D	58	A
9	A	19	A	29	A	39	D	49	C	59	C
10	D	20	C	30	B	40	B	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) One crore = 1,00,00,000 = a 1 followed by 7 zeroes (10^7). (A lakh has 5 zeroes; an arab has 9 — both are tempting near-misses.)

2. (A) 50000000 has 8 digits. Indian grouping is 3-2-2-2 from the right: 000 | 00 | 00 | 5 → 5,00,00,000 (five crore). Option D (50,000,000) is the International 3-3-3 grouping, and B/C have the wrong digit count.

3. (D) One million = 1,000,000 = 10,00,000 = **10 lakh**. The classic trap is treating a million and a lakh as equal — a million is ten times a lakh.

4. (B) Adding 1 to the largest 5-digit number: $99,999 + 1 = 1,00,000$, the smallest 6-digit number (one lakh). 10,000 is only the smallest 5-digit number.

5. (A) Group 7,84,36,512 as 7 crore | 84 lakh | 36 thousand | 512. The digit 4 sits in the lakh place of "84 lakh," so it stands for **4 lakh**. (40 lakh would be the 8.)

6. (D) Three crore = 3,00,00,000; five lakh = 5,00,000; seven thousand = 7,000; two hundred = 200. Sum = 3,05,07,200. Option C (3,05,70,200) misplaces the thousand digit;

B inserts an extra lakh.

7. (B) Strip the commas: 90706005. Re-group International 3-3-3 from the right: 005 | 706 | 90 → 90,706,005. The digit string is identical to the Indian 9,07,06,005 — only the comma positions change.

8. (C) The smallest 8-digit number is 1 followed by seven zeroes = 1,00,00,000 (one crore). Option B (10,00,00,000) is 9 digits; A is 7 digits.

9. (A) In 6,53,000 the 6 is in the lakh place → place value 6,00,000; the 3 is in the thousand place → 3,000. Difference = 6,00,000 – 3,000 = 5,97,000. (Subtracting bare digits 6 – 3 = 3 gives the trap 3,00,000.)

10. (D) The Indian convention puts the first comma after 3 digits from the right, then after every 2 digits: **3, then 2, then 2, then 2, ...** The 3-3-3 pattern (option A) is the International system.

11. (C) One arab = 100 crore, and one crore = 100 lakh, so one arab = $100 \times 100 =$ **10,000 lakh**. (100 lakh is only one crore.)

12. (B) One billion = 10^9 = 100 crore = 100×100 lakh = **10,000 lakh**. Estu's "one hundred lakh" equals only one crore, off by a factor of 100.

13. (A) Convert all to plain numbers: 9 lakh = 9,00,000; 1 million = 10,00,000; 90 lakh = 90,00,000; and 9,00,001. The smallest is **9 lakh** = 9,00,000 (just one less than 9,00,001).

14. (B) $4 \times 10,00,000 = 40,00,000$; $7 \times 10,000 = 70,000$; $3 \times 100 = 300$; plus 5. Sum = 40,70,305. Note the hundred-thousands and ten-thousands places, with zeros elsewhere.

15. (B) Starting from 0, each +1000 press adds 1000. To reach 53,000 you need $53,000 \div 1000 =$ **53** presses.

16. (A) Each +100 press adds 100. One lakh = 1,00,000, so $1,00,000 \div 100 =$ **1,000** presses. (This also shows 1,000 hundreds make a lakh.)

17. (C) Fewest presses = place-value digits of 40,629: $4 + 0 + 6 + 2 + 9 =$ **21**. The press-counts in each slot are exactly the digits.

18. (D) $8,300 - (7 \times 1000) = 8,300 - 7,000 = 1,300$, which needs $1,300 \div 100 =$ **13** presses of +100. (This is the valid non-standard decomposition $7 \times 1000 + 13 \times 100$.)

19. (A) 3,07,250 has place-value digits (lakh → ones): 3, 0, 7, 2, 5, 0. The minimal press-count in each slot equals these digits, so **3, 0, 7, 2, 5, 0**. Option B drops the zero in the ten-thousand slot.

- 20. (C)** Check each: $A = 5000 + 70 + 2 = 5072$ ✓; $B = 5000 + 72 = 5072$ ✓; $D = 5000 + 60 + 12 = 5072$ ✓; $C = 4000 + 1000 + 70 = 5070 \neq 5072$. So C does NOT make 5,072.
- 21. (D)** Group 12,78,830 as 12 lakh | 78 thousand | 830 → **twelve lakh seventy-eight thousand eight hundred thirty**. Option C wrongly says "eight hundred three"; A and B regroup the digits incorrectly.
- 22. (A)** 6,72,85,183 = 6 crore | 72 lakh | 85 thousand | 183. In "72 lakh," the 7 occupies the ten-lakh place, so the answer is 7. (The 2 is the lakh place.)
- 23. (C)** In 5,35,000 the first 5 is in the lakh place (place value 5,00,000); the second 5 is in the thousand place (place value 5,000). Difference = $5,00,000 - 5,000 = 4,95,000$.
- 24. (B)** "Ten crore twenty lakh thirty thousand forty" = 10,20,30,040. Strip commas: 102030040. Re-group International 3-3-3: 040 | 030 | 102 → **102,030,040**.
- 25. (B)** 10,00,000 (Indian) and 1,000,000 (International) are the **same number, ten lakh** (= one million). The comma positions and spoken names differ, but the value does not. Options A and C wrongly claim one is ten times the other.
- 26. (A)** One lakh = 1,00,000. So $1,00,000 - 75,000 = 25,000$. (15,000 would be the gap to 90,000; 1,25,000 wrongly adds instead of subtracting.)
- 27. (D)** Nearest ten-thousand of 4,87,650: the thousands portion is 7,650, which is $\geq 5,000$, so round the ten-thousands place up: 4,80,000 → 4,90,000. Answer **4,90,000**.
- 28. (C)** Nearest thousand of 3,87,69,957: look at the last 3 digits, $957 \geq 500$, so round up: 3,87,69,000 → **3,87,70,000**.
- 29. (A)** Total needed = $732 + 23 = 755$, and she must never run short, so round **up to 800**. Ordering exactly 755 leaves no margin; rounding down (700 or 750) risks shortage.
- 30. (B)** Quoting ₹470 as "about ₹450" lowers the figure, so it is rounding **down**, done to seem generous (and because customers like round numbers). It is the nearest fifty, not a "round up."
- 31. (C)** Nearest lakh of 6,72,85,183: the last 5 digits form 85,183, which is $\geq 50,000$, so round the lakhs up: 6,72,00,000 → **6,73,00,000**.
- 32. (D)** $4,63,128 + 4,19,682 = 8,82,810$ — near 8,00,000 and just above it, confirming Roxie. Answer **8,82,810**. (8,72,810 drops a carry; 9,82,810 over-carries.)
- 33. (A)** A round "about one lakh" **conveys the size with less effort and no real loss** of meaning — the exact 1,02,455 isn't needed at the dinner table. The exact figure isn't "wrong"; rounding is a deliberate, honest approximation.

34. (B) Nearest crore of 8,49,21,000: the part below the crore is 49,21,000, which is $< 50,00,000$, so round **down: 8,00,00,000**. (Reading 49,21,000 as "almost a crore" and rounding up to 9 crore is the trap.)

35. (C) Numbers that round to 7,00,000 at the nearest lakh lie in $[6,50,000, 7,49,999]$. Checking: $7,51,000 \rightarrow 8$ lakh; $6,40,000 \rightarrow 6$ lakh; $7,62,000 \rightarrow 8$ lakh; only **7,48,200** rounds to 7,00,000.

36. (B) $116 \times 5 = 116 \times (10 \div 2) = (116 \times 10) \div 2 = 1160 \div 2 = \mathbf{580}$. (Forgetting to halve gives the trap 1,160.)

37. (D) $824 \times 25 = 824 \times (100 \div 4) = 82400 \div 4 = \mathbf{20,600}$. (Stopping at 82,400 forgets the $\div 4$; 8,240 divides by 10 instead.)

38. (A) $72 \times 125 = 72 \times (1000 \div 8) = 72000 \div 8 = \mathbf{9,000}$. (576 is just 72×8 , the wrong operation.)

39. (D) A 3-digit $\times$ 3-digit product has $(3+3-1)$ or $(3+3) = \mathbf{5 \text{ or } 6}$ digits. Smallest $100 \times 100 = 10,000$ (5 digits); largest $999 \times 999 = 998,001$ (6 digits). Indeed $234 \times 567 = 1,32,678$ has 6 digits — within the range.

40. (B) 2-digit $\times$ 2-digit: smallest $10 \times 10 = 100$ (3 digits), largest $99 \times 99 = 9,801$ (4 digits). So the product has **3 or 4 digits** — never exactly 4 only.

41. (C) The smallest 3-digit $\times$ 3-digit product uses the smallest inputs: $100 \times 100 = \mathbf{10,000}$. (9,801 is 99×99 , the largest 2-digit case — a tempting mix-up.)

42. (D) m-digit $\times$ n-digit gives $(m+n-1)$ or $(m+n)$ digits. Here $12 + 13 = 25$, so the answer has **24 or 25** digits. (156 wrongly multiplies 12×13 .)

43. (B) $8,000 \times 7,000 = 8 \times 7 \times 1000 \times 1000 = 56 \times 10^6 = 56,000,000$ — which has **8** digits. (Using the m+n rule: $4 + 4 = 8$ here.)

44. (D) Roxie is **wrong; the smallest such product is 10,000**. The least 3-digit $\times$ 3-digit product is $100 \times 100 = 10,000$ (already 5 digits), so a 4-digit result is impossible. (Options A and B illegally use a 2-digit or 1-digit factor.)

45. (A) The shortcut works **because 5 is half of 10**: multiplying by 5 equals multiplying by 10 (easy) then halving. Option C is false ($\times 10$ makes a number ten times bigger), and dividing by 2 is not the same as multiplying by 2.

46. (C) 1 lakh buses $\times$ 50 people each $= 1,00,000 \times 50 = 50,00,000 = \mathbf{50 \text{ lakh}}$. (5 lakh divides instead of placing the zero; 5 crore over-counts by ten.)

47. (B) Seconds in a day $= 24 \times 60 \times 60 = 86,400$. At one number per second you reach **about 86,400** — far short of a million. (One lakh, 1,00,000, is close but still more than

86,400.)

48. (D) $100 \text{ km/day} \times 365 \text{ days} \times 10 \text{ years} = 3,65,000 \text{ km}$. The Moon is 3,84,400 km away, so she falls about 19,400 km short — **3,65,000 km — just short**.

49. (C) $5,000 \text{ ships} \times 2,500 \text{ passengers} = 5000 \times 2500 = 1,25,00,000$ (1 crore 25 lakh). Answer **1,25,00,000**. (12,50,000 drops a zero.)

50. (D) $75 \text{ beats/min} \times 60 \text{ min} \times 24 \text{ h} = 1,08,000 \approx \text{about } 1,00,000$ (one lakh) beats a day. (10,000 and 4,500 are far too small; 10,00,000 is ten times too big.)

51. (A) A reasonable Fermi assumption matches everyday experience: **one barbershop serves about 5,000 people in a locality**. "1,000 haircuts a day" is impossible (each takes ~30 min); a bus holding 5,000 people and a city with exactly one barbershop are absurd.

52. (B) 500 lakhs = 5,00,00,000 = 5 crore = 50 million — which is far more than 5 million. So **500 lakhs is bigger (it is 5 crore)**. They are not equal; the trap is reading "500 lakh" as "5 million."

53. (C) $640 \text{ crore} = 640 \times 10^7 = 6.4 \times 10^9 = 6.4 \text{ billion}$, which is less than 60 billion. So **60 billion (640 crore is only 6.4 billion)** is bigger. (Reading 640 crore as 64 billion mis-scales by a factor of ten.)

54. (A) As plain numbers: 90,000; 9 lakh = 9,00,000; 1 million = 10,00,000; 1 crore = 1,00,00,000. Increasing order: **90,000 < 9 lakh < 1 million < 1 crore**.

55. (C) One crore = 1,00,00,000. Increase = $1,00,00,000 - 5,00,000 = 95,00,000$ **litres**. (50,00,000 would be the gap from 50,00,000, not 5,00,000.)

56. (B) The smallest 9-digit number is 1 followed by eight zeroes = 10,00,00,000; the number just before it is 9,99,99,999 (the largest 8-digit number). So **10,00,00,000 and 9,99,99,999**.

57. (D) 12 lakh = 12,00,000 and 8 crore = 8,00,00,000. Fraction = $12,00,000 / 8,00,00,000 = 12 / 800 = 3/200$. ($1/80 = 10,00,000 / 8,00,00,000$, a near-miss from rounding 12 to 10.)

58. (A) 1 lakh buses $\times 50 = 50,00,000 = 50 \text{ lakh capacity}$. Mumbai $\approx 1 \text{ crore } 24 \text{ lakh} = 124 \text{ lakh people}$. $50 / 124 \approx 0.40$, so **about 40%** — the buses hold only part of the city.

59. (C) Compare: 1,00,01,000 (just over one crore) vs 99,99,000 (under one crore) $\rightarrow 1,00,01,000 > 99,99,000 \checkmark$. The others are false: $7,05,000 < 70,50,000$; $9,99,999 < 10,00,000$; $8 \text{ lakh} < 8 \text{ million}$.

60. (D) 49,873 to the nearest ten-thousand: the thousands portion $9,873 \geq 5,000$, so round up $\rightarrow 50,000$. But "about a lakh" (1,00,000) is roughly double the true $\sim 50,000$, so **nearest ten-thousand is 50,000; "about a lakh" overstates by roughly double.**

Solutions complete: 60 / 60.